

How to Protect Telecom Networks

Automatic protection switching schemes allow configuring a pair of lines for line redundancy. When the working interface fails, the protection interface quickly assumes the network traffic and provides end users with a continuous operation. Let's delve deep into protection mechanisms...

DR RAJIV KUMAR SINGH

Network failures, whether due to human error or faulty technology, can be very expensive for users and telecom service providers alike. As a result, the subject of so-called fall-back mechanism is currently one of the most talked about in the telecom world. A wide range of standardised mechanisms is incorporated into synchronous networks in order to compensate for failures in network path/elements and to provide highly-available telecom networks.

Automatic protection switching

Automatic protection switching (APS) is a fault-tolerant topology that is used for providing backup to telecom networks. For network survivability, in the event of failure in a network element or link, APS involves reserving a protection channel with the same capacity as the channel or facility to be protected. In the event of signal-fail (SF) or signal-degrade (SD) condition, the working line switches automatically to the protection line within a few milliseconds.

Fig. 1 shows basic APS configuration on network elements, for example, routers/nodes A and B. Here, node A is configured with the working interface and node B is configured with protection interface. In a router configured for APS, configuration for the protection interface includes the IP address of the router (normally its loopback address) that has the working interface. Normally, the working and protection interfaces are connected to a network element of the transmission system, typically, an add-drop multi-

plexer (ADM).

In the event of failure on working interface of node A, the connection automatically switches over to the protection interface on node B. On the protection circuit, K1 and K2 bytes from the line overhead (LOH) of the

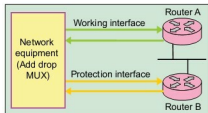


Fig. 1: Basic APS architecture

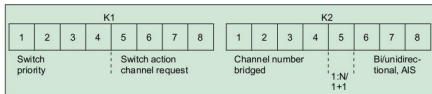


Fig. 2: The K1, K2 bytes

synchronous digital hierarchy (SDH) frame indicate the current status of the APS connection and convey any requests for action.

K1 and K2 bytes. The K1 byte in SDH configuration contains switching pre-emption priorities (in bits 1 to 4) and channel number of the channel requesting action (in bits 5 to 8). The K2 byte contains channel number of the channel that is bridged onto protection (bits 1 to 4) and mode type (bit 5); besides, bits 6 to 8 contain various conditions, such as multiplex-section alarm-indication signal (MS-AIS) and indication of unidirectional or bidirectional switching (Fig. 2).

The APS is very extensible in terms of topologies (for example, rings) and flexibility (for example, link-capacity-

adjustment scheme (LCAS) service restoration). Basically, two types of protection architectures—linear protection and ring protection—are distinguished in APS. The linear-protection mechanism is adopted for point-to-point connections. But ring-protection mechanism can take on many different forms. Both mechanisms use spare circuits or components to provide the back-up path.

Linear protection

1+1 APS architecture. The simplest form of mechanism for network survivability in the event of network fail-

ure is 1+1 APS. Here, each and every working transmission path/line/channel is protected by one protection path/line/channel (Fig. 3). At the near end, the signal is bridged permanently, that is, split into two identical signals, and sent over both the working and the protection lines simultaneously. At the far end, signal selection is made on the basis of switch initiation/trigger criteria, which are signal fail (SF), signal degrade (SD), loss of signal (LOS) or loss of frame (LOF).

If a defect occurs, the protection agent/switch in the network elements at both ends switches the circuit over to the protection line. Switching at the far end is initiated by the return of an acknowledgment in the backward channel. 1+1 architecture includes